Phase-Space Volume Contraction, Not the 0–1 Test, Separates Conservative from Dissipative Chaos: A Cross-System Cautionary Result

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Abstract

Deterministic chaos defeats long-horizon trajectory prediction, but many structural features of a chaotic system remain measurable. We test whether a scalar diagnostic can distinguish conservative Hamiltonian chaos from dissipative chaos, rather than merely detecting chaos itself. Across a 26-system benchmark, an earlier version of this study appeared to show that the Gottwald–Melbourne 0–1 statistic separates conservative from dissipative systems. A trajectory-length audit falsified that conclusion: for chaotic Hamiltonian systems the statistic is length-dependent and can climb from near zero to near one (double pendulum: $K \approx -0.13$ at 4k steps to $K \approx 0.99$ at 16k steps, mean over 4 initial conditions). The 0–1 statistic is therefore a chaos detector, not a conservative/dissipative classifier. The robust discriminator is instead the phase-space volume-contraction rate $\Sigma\lambda = \langle \nabla \cdot f \rangle$, equal to the sum of Lyapunov exponents: zero to within 10^{-3} for all 8 Hamiltonian systems and strictly negative for all 11 finite-dimensional dissipative flows, matching analytic traces where available. We show why the false 0–1 split is seductive: the entire K-vs-length curve is predicted from the observable autocovariance by a closed-form Green–Kubo expression (Pearson $r = 0.955$ against measured K). A held-out symbolic-itinerary experiment further shows that attractor transition grammar is predictable (0.770 vs. 0.233 baseline on cell changes, 21/23 chaotic systems beating baseline) even when microstate forecasts are Lyapunov-limited. A battery of corroborating diagnostics—the M_c growth exponent, autocorrelation-time observable selection, the correlation decay law, an effective run-length calculator, and order-k route grammar—independently converges on the same reading: each is set by the observable's coarse-grained mixing rate, not by conservation structure. The contribution is a corrected discriminator, a reproducible warning about finite-time misuse of a widely used chaos statistic [7], and a bounded constructive result about predicting the climate, not the weather.

1 Introduction

Chaotic systems share sensitivity to initial conditions, yet they do not share the same geometric or physical structure. A Hamiltonian double pendulum preserves phase-space volume and remains on an energy shell; the Lorenz system [1] contracts onto a strange attractor. A scalar chaos detector can report that both systems are chaotic without answering whether the chaos is conservative or dissipative.

This paper asks a narrow question: among low-cost diagnostics—those requiring only a scalar time series, not the full Lyapunov spectrum or the model vector field—what separates conservative Hamiltonian chaos from dissipative chaos? We tested two candidates. The first was the Gottwald–Melbourne 0–1 test [5, 6, 7], which maps a scalar time series to a statistic K approaching one for chaotic dynamics and zero for regular dynamics. The second was phase-space volume contraction, $\Sigma\lambda = \langle \nabla \cdot f \rangle$, the divergence of the vector field averaged along the trajectory, equal to the sum of Lyapunov exponents for smooth flows [8, 9]. Liouville's theorem makes the latter a natural discriminator: Hamiltonian flows preserve phase-space volume, while dissipative flows contract it [4].

The central result is a negative one. Early runs at short trajectory lengths suggested that conservative chaos had $K \approx 0$ while dissipative chaos had $K \approx 1$. Longer trajectories and initial-condition ensembles overturned that reading. We therefore present the correction as part of the result: the 0–1 test remains useful as a chaos detector, but it should not be used as a conservative/dissipative classifier without additional structure. Five further low-cost diagnostics (Secs. 3.5–3.9) reinforce the correction from independent angles—growth-rate exponent, observable selection, itinerary advantage, decay law, and route-grammar memory—each governed by the observable's coarse-grained mixing rate rather than by conservation.

2 Systems and measurements

The benchmark contains 26 systems spanning maps, dissipative flows, Hamiltonian flows, delay dynamics, and non-chaotic controls (Table 1). Systems include the Lorenz 63 flow [1], the Hénon–Heiles potential [2], and the Mackey–Glass delay equation [3], among others listed in Table 1. Flows are integrated by fixed-step RK4 after warm-up onto the attractor; maps are iterated directly. Mechanical Hamiltonian systems are integrated in canonical (q, p) coordinates. Known analytic invariants are tracked as integration hygiene checks.

0–1 test. For a scalar observable $\phi(j)$ and frequency c, define translation variables

$$p_c(n) = \sum_{j=1}^{n} \phi(j)\cos(jc), \qquad q_c(n) = \sum_{j=1}^{n} \phi(j)\sin(jc). \tag{1}$$

The mean-square displacement over lag n is

$$M_c(n) = \frac{1}{N}\sum_{j=1}^{N}\big[(p_c(j+n) - p_c(j))^2 + (q_c(j+n) - q_c(j))^2\big]. \tag{2}$$

We use $K_c = \mathrm{corr}(\log n, \log M_c(n))$, averaged over $n_c = 12$ random frequencies c uniformly drawn from $(0, \pi)$. This log–log form is equivalent to the linear-regression coefficient used by Gottwald and Melbourne [6] in the asymptotic regime where $M_c(n) \sim n$ for chaotic and $M_c(n) \sim \text{const}$ for regular dynamics, but it can assign high K to quasiperiodic orbits near resonance (where $M_c(n) \sim n^2$); this point is noted in Section 5. In the asymptotic regime, $K \to 1$ for chaotic dynamics and $K \to 0$ for regular dynamics. The scalar observable is the first coordinate of the state vector for flows; for maps the natural scalar output is used. The finite-length regime is the failure mode studied here.

Phase-space volume contraction. For a vector field $\dot{x} = f(x)$,

$$\Sigma\lambda = \langle \nabla \cdot f(x(t)) \rangle = \left\langle \sum_i \frac{\partial f_i}{\partial x_i} \right\rangle. \tag{3}$$

For smooth flows this equals the sum of Lyapunov exponents. Hamiltonian flows have $\nabla \cdot f = 0$ in canonical coordinates; dissipative flows contract volume. Where analytic traces are available, they are used as cross-checks against the numerical central-difference estimator.

3 Results

3.1 The 0–1 statistic is not a conservative/dissipative discriminator

On sufficiently long trajectories, Hamiltonian K values scatter across the same range as dissipative chaos. The decisive evidence is the length audit (Fig. 1). Dissipative systems such as Lorenz and Chen read $K \approx 1$ at all tested lengths. Hamiltonian systems can begin near zero and climb toward one as the trajectory length grows. The double pendulum moves from $K \approx -0.13$ (mean over 4 initial conditions) at 4k steps to $K \approx 0.99$ at 16k steps; the coupled Duffing Hamiltonian moves from $K \approx -0.14$ at 4k to $K \approx 0.94$ at 16k. This is the pattern one expects from a chaos detector whose finite-time behavior is governed by observable mixing time, not by whether the system is conservative (Section 3.3).

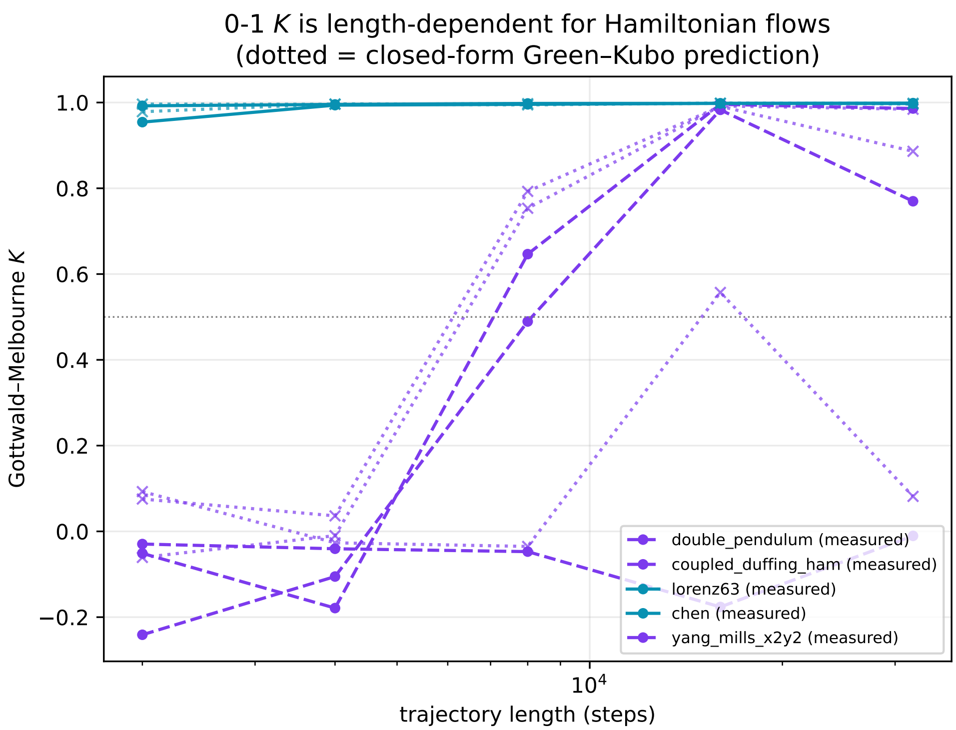

Figure 1: The 0–1 statistic is trajectory-length dependent for slow-mixing Hamiltonian systems. Purple curves can climb from near zero to near one; cyan dissipative systems read near one at all lengths. Dotted curves are the closed-form Green–Kubo predictions.

3.2 Volume contraction separates the classes

The phase-space contraction rate cleanly separates the finite-dimensional systems with defined vector fields (Fig. 2). The three maps and the Mackey–Glass delay system are excluded from this analysis because they lack a finite-dimensional smooth vector field; $\Sigma\lambda$ is defined only for ordinary differential equation systems. Among those, all 8 Hamiltonian flows have $\Sigma\lambda = 0$ to within 10^{-3}, and all 11 dissipative flows have $\Sigma\lambda < 0$. The estimator reproduces analytic traces where they exist: Lorenz63 gives $-13.667 = -\sigma - 1 - \beta$, Chen gives -10, Lorenz96$_N$ gives $-N$, Sprott B gives -1, Thomas gives $-3b$, and Ueda gives $-k$. Because $\Sigma\lambda$ is a vector-field property rather than a finite scalar-series random walk, it has no trajectory-length crossover analogous to the 0–1 statistic.

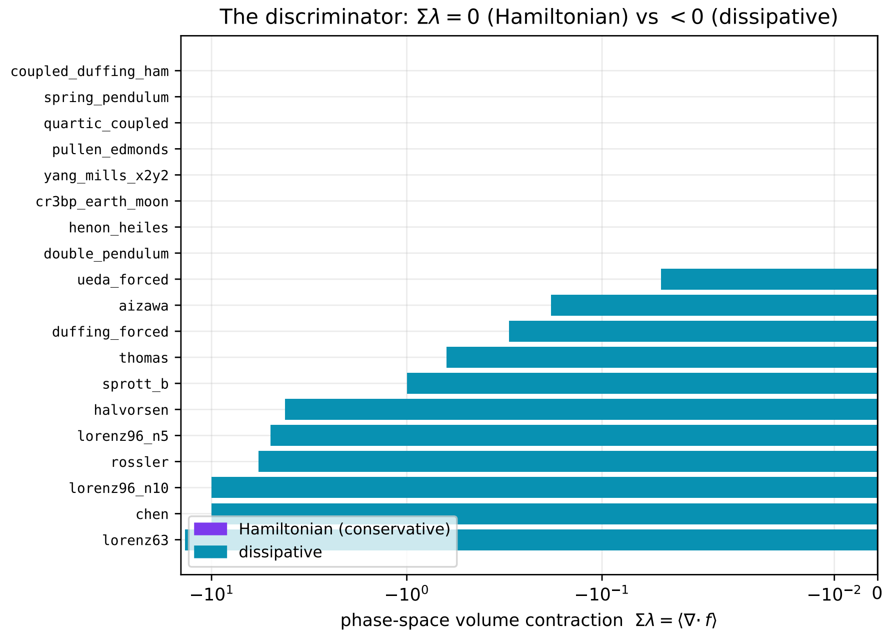

Figure 2: Phase-space volume contraction separates conservative from dissipative chaos. Hamiltonian systems sit at zero; dissipative flows contract. The horizontal axis is symlog so both weak and strong contraction are visible.

The Hamiltonian cases also pass integration hygiene: all 8 systems conserve an analytic invariant (energy, or the Jacobi integral for CR3BP), and the relative drift in that invariant is at most 7×10^{-4} over the run. The measured largest Lyapunov exponent is positive for every chaotic Hamiltonian system. Thus the zero volume contraction is not a failure to reach chaos; it is the expected Liouville structure.

3.3 Why the 0–1 failure happens

The finite-time behavior of K is not mysterious. For a mean-zero stationary observable ϕ (mean-subtracted in the implementation) with autocovariance $C(k) = E[\phi(j)\phi(j+k)]$, expanding Eq. (2) and applying stationarity yields the exact mean-square displacement

$$M_c(n) = \sum_{k=-(n-1)}^{n-1} (n-|k|)C(k)\cos(kc) = nC(0) + 2\sum_{k=1}^{n-1}(n-k)C(k)\cos(kc). \tag{4}$$

Its asymptotic diffusion coefficient is the Green–Kubo / Wiener–Khinchin power spectral density at frequency c [10, 11]:

$$D_c = \lim_{n\to\infty} \frac{M_c(n)}{n} = C(0) + 2\sum_{k\geq 1} C(k)\cos(kc). \tag{5}$$

Since K_c is the log–log correlation computed from $M_c(n)$, the observable autocovariance predicts the full length-dependent curve. In the artifacts, the predicted K from Eq. (4) matches measured K with Pearson $r = 0.955$ and mean absolute error 0.077 (Fig. 3); the predicted crossover length is within one ladder step on 20/23 chaotic systems. The apparent class split was therefore a mixing-time artifact.

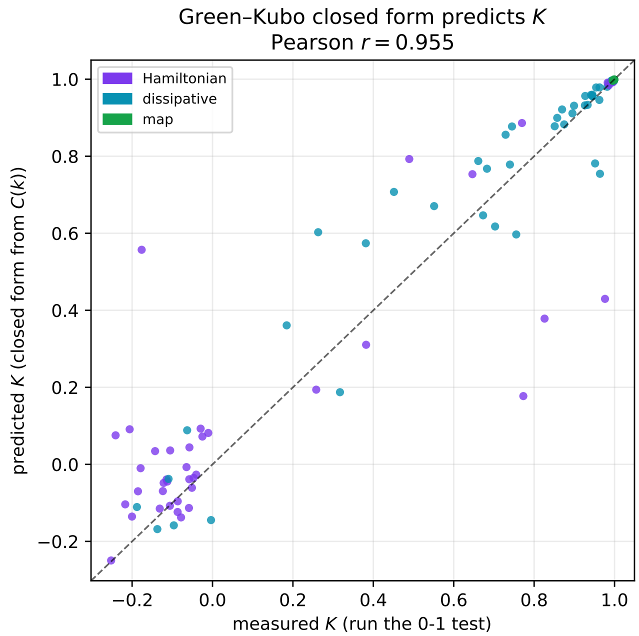

Figure 3: Closed-form Green–Kubo prediction of measured 0–1 K from the observable autocovariance alone.

3.4 Held-out itinerary grammar: predict the climate, not the weather

The same observatory also measures symbolic attractor grammar. Each attractor is coarsened into $k = 6$ cells by k-means on the first half of the trajectory (results are robust to $k \in \{4, 6, 8\}$; all values are in the data artifacts); an order-1 Markov model is then evaluated on the held-out second half. Raw next-symbol accuracy across 23 chaotic systems is 0.925, but most of that is trivial dwell-time persistence: finely sampled flows tend to remain in the same cell, and the persistence baseline is 0.863. The non-trivial result conditions on the roughly 14% of steps where the cell changes. On those transitions, the learned grammar predicts the destination cell with mean accuracy 0.770 versus a 0.233 baseline, beating baseline on 21/23 systems (Fig. 4); the remaining 2 (Duffing forced, Ueda forced) had zero cell transitions in the held-out half at $k = 6$ and are excluded from the transition mean. This is not microstate prediction. It is a bounded, held-out statement that chaotic attractors can have learnable symbolic route grammar.

3.5 Growth-rate diagnostics: M_c slope analysis

The log–log correlation K_c collapses the entire $M_c(n)$ trajectory to a single scalar. A complementary view is the *growth-rate exponent* $\alpha = \mathrm{d}\log M_c/\mathrm{d}\log n$, estimated by OLS on the c-averaged $\log M_c(n)$ over $N_{\text{lag}} = 30$ geometrically-spaced lags (data: `results/chaos_slope_analysis.json`). All trajectories use $N = 12{,}000$ integration steps—three times the 4,000-step primary runs of Table 1—to expose asymptotic behaviour, with the same decimation procedure as Section 3.3. The long-range sub-slope α_{lo} ($n \geq N_{\text{dec}}/2$) additionally characterises whether growth saturates at longer lags.

α–K decoupling, and the fragility of K. Two Hamiltonian systems show an α–K decoupling that survives frequency resampling: CR3BP ($\alpha = 0.847$, $K = -0.05$) and quartic coupled ($\alpha =$

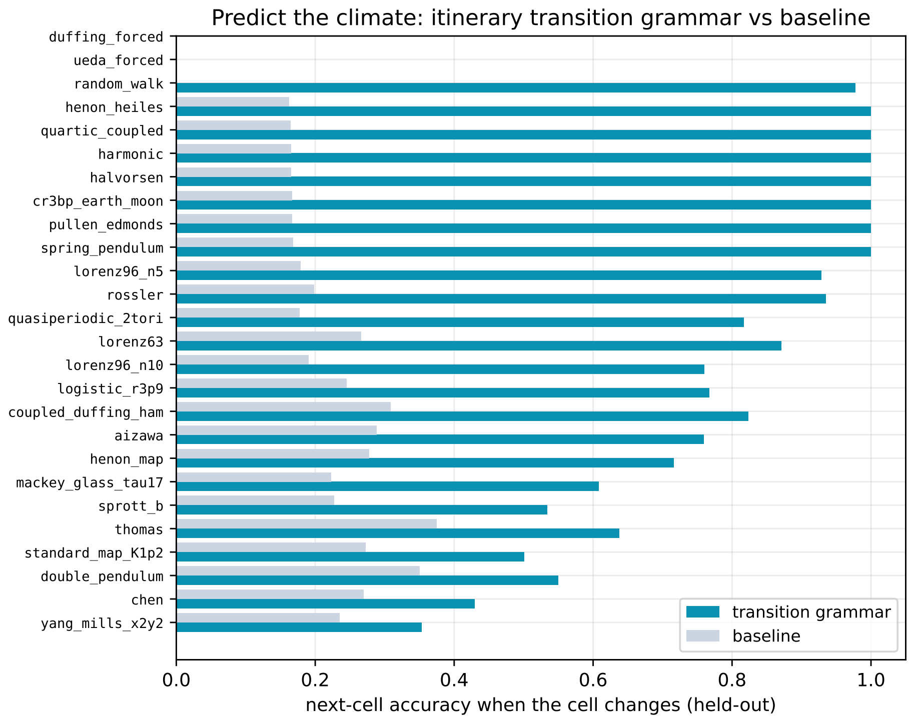

Figure 4: Held-out next-cell prediction when the cell changes. The transition grammar beats the most-common-transition baseline on 21/23 chaotic systems.

0.888, $K = -0.11$) grow M_c at slope near one yet keep a median K below zero, with 83% and 92% of 12-frequency draws falling below 0.3 (Fig. 5; Table 2, bold). Pullen–Edmonds also holds K low (-0.05, 92% below 0.3) but its M_c partially saturates ($\alpha_{\mathrm{lo}} = -0.15$), so its growth is not cleanly diffusive. The mechanism is direct: α is estimated from the mean $M_c(n)$ averaged over all frequencies, which smooths per-frequency noise, whereas K_c is the *median* per-c log–log correlation; for slow-mixing Hamiltonians with competing KAM structure the per-c curves are noisy and non-monotonic, holding the median near or below zero even though the mean M_c grows with slope near one.

This same per-c variance makes K itself fragile for these systems. Re-drawing the 12 frequencies moves spring pendulum's median K across $[-0.12, +0.96]$ at fixed length and trajectory; its typical value is $+0.32$ (only half the draws below 0.3), so a single negative reading is a sampling artifact, not a stable decoupling—indeed the main-lab 0–1 implementation returns $K \approx +0.98$ for the same spring-pendulum trajectory at $N = 12{,}000$. CR3BP and quartic are not immune (their draws exceed $+0.9$ in places too) but stay low in the large majority of draws, which is why we treat only them as robust. Yang–Mills is separate, with $\alpha = -0.08$ (bounded M_c at this length—neither diffusive nor a decoupling); Hénon–Heiles ($\alpha = 0.95$, $K = 0.58$), double pendulum ($\alpha = 1.31$, $K = 0.71$) and coupled Duffing ($\alpha = 0.76$, $K = 0.94$) keep both measures positive. The lesson reinforces the central thesis: for slow-mixing systems the 0–1 K is a finite-sampling estimate sensitive to the frequency draw, not a stable invariant.

Quasiperiodic control and the sub-slope. The non-chaotic controls expose the same frequency fragility. The main-lab 0–1 reads $K = 0.961$ for the quasiperiodic 2-torus (Table 1), yet on the same 12,000-step trajectory the slope pipeline's frequency-median is only $+0.10$ (range

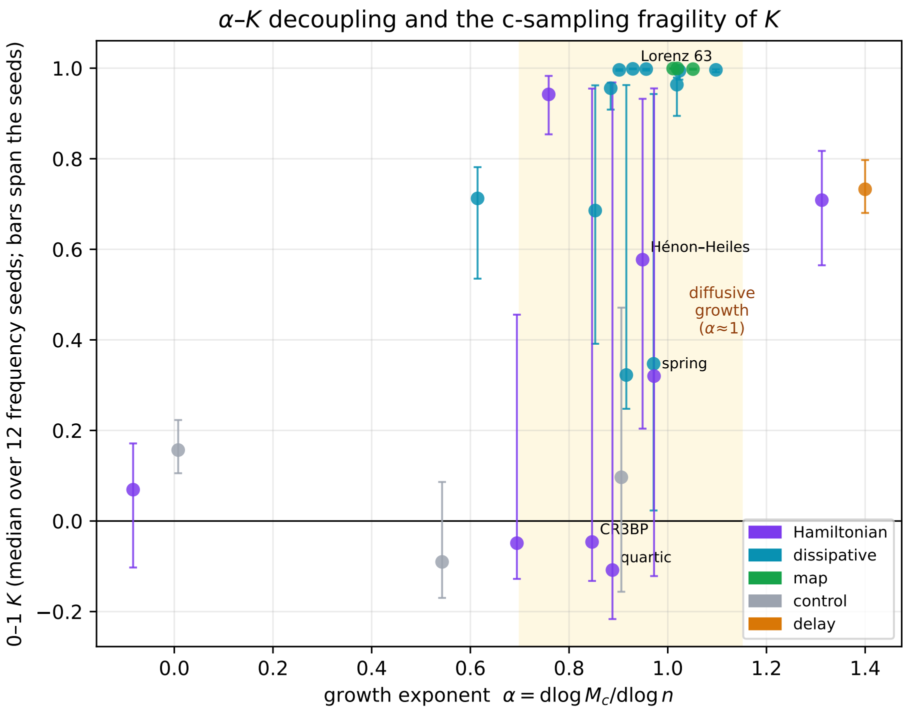

Figure 5: Growth exponent α versus the 0–1 K (median over 12 random frequency seeds; error bars span the seeds) at $N = 12{,}000$ steps. CR3BP and quartic coupled occupy the decoupling corner—$\alpha \approx 1$ yet $K < 0$ robustly. The long Hamiltonian error bars are the c-sampling fragility: a single frequency draw can place K anywhere from -0.2 to $+1$ (spring pendulum), whereas fast-mixing dissipative flows and maps pin $K \approx 1$ with negligible spread.

$[-0.16, +0.47]$, $\alpha = 0.91$): a bounded, non-chaotic orbit whose 0–1 value swings with the frequency draw, exactly as the slow Hamiltonians do. Here the long-range sub-slope is decisive where K is not: $\alpha_{\mathrm{lo}} = +0.37$ (partial saturation) for the 2-torus versus -0.81 for the harmonic oscillator (unambiguous saturation) flags the bounded character that a single K reading can miss. Neither α alone nor K alone reliably separates quasiperiodic from slow-mixing chaotic at finite trajectory lengths.

3.6 Observable selection via autocorrelation screening

For multi-dimensional systems, the 0–1 test requires selecting a scalar observable ϕ. All primary results (Table 1) use the first state coordinate, but other coordinates can converge K faster or slower. The Green–Kubo framework predicts the leading trend: a coordinate with shorter integrated autocorrelation time $\tau_{\mathrm{int}} = \sum_{k\geq 0} C(k)/C(0)$ reaches the asymptotic $M_c(n) \sim n$ regime sooner, giving a smaller crossover length $n_{1/2}$.

The Rössler system makes this concrete (Fig. 6). Using the same trajectory and c-values, the z coordinate ($\tau_{\mathrm{int}} = 7.5$ steps) exceeds $K = 0.5$ at $N = 1{,}000$ steps, while y ($\tau_{\mathrm{int}} = 12$ steps) does not cross until $N = 8{,}000$ steps—an 8-fold speedup from coordinate choice, with τ_{int} correctly predicting the ordering. The physical reason is structural: Rössler z captures the rare, rapidly-decorrelating spike events whose autocorrelation decays within tens of steps, whereas x and y record the slow spiral that takes many orbits to decorrelate, keeping $M_c(n)$ pre-asymptotic and depressing K at

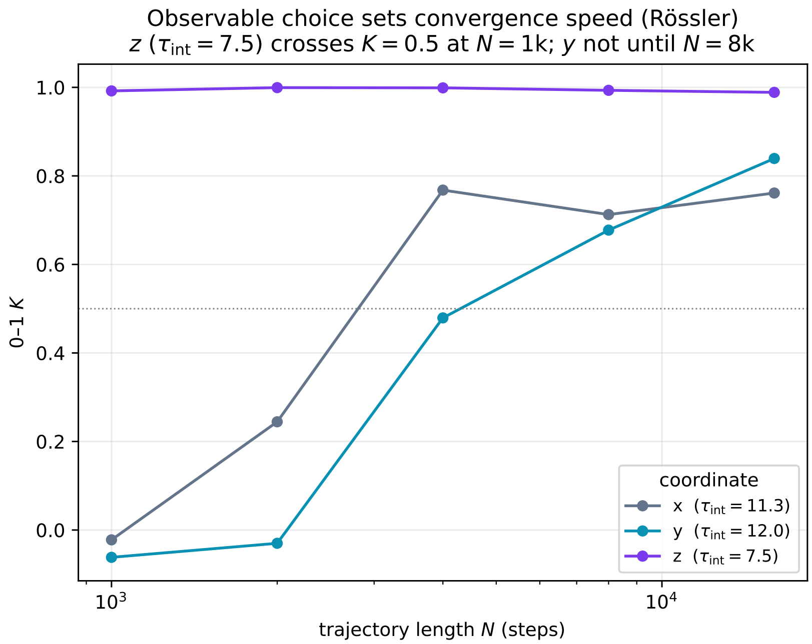

Figure 6: Observable selection for the Rössler system: the 0–1 K versus trajectory length for each state coordinate. The z coordinate ($\tau_{\mathrm{int}} = 7.5$) is already converged ($K \approx 1$) at $N = 1{,}000$ steps, whereas y ($\tau_{\mathrm{int}} = 12$) does not cross $K = 0.5$ until $N = 8{,}000$. Shorter integrated autocorrelation time gives faster convergence.

short trajectory lengths.

The relationship is directional but imperfect. For the double pendulum, θ_2 ($\tau_{\mathrm{int}} = 239$ steps) never converges within 16,000 steps, correctly identified as the worst choice by τ_{int} screening. However, p_2 ($\tau_{\mathrm{int}} = 84$ steps) converges *faster* than θ_1 ($\tau_{\mathrm{int}} = 48$ steps) because its K curve rises monotonically whereas θ_1's oscillates—dipping to $K = -0.15$ near $N = 4{,}000$—before converging. Minimum τ_{int} screens out catastrophically slow coordinates reliably; it does not guarantee the global minimum crossover length. Full per-coordinate convergence curves are in `results/chaos_observable_selection.json`.

3.7 Itinerary predictability tracks mixing, not conservation

Does the held-out itinerary grammar (Fig. 4) separate the two dynamical classes? We score each system by its genuine transition advantage $\mathrm{adv} = a_{\mathrm{trans}} - a_{\mathrm{base}}$, the gain of the order-1 move grammar over the most-common-destination baseline, evaluated only on the steps where the coarse cell actually changes. (The order-1-minus-order-0 lift is excluded; it is dominated by dwell-time persistence.) We require at least 100 cell changes per system, so the binomial standard error on a_{trans} is at most 0.05, and join the result to the largest (Benettin) Lyapunov exponent λ_1 (Table 1) and the lab's 12,000-step 0–1 K_{median}.

At face value the Hamiltonians win: on the reliable subset $\overline{\mathrm{adv}} = 0.835$ (four systems) against 0.564 for the dissipative flows (eight systems), Mann–Whitney $p = 0.022$, with a positive gap at every cell count $K \in \{4, 6, 8\}$ (gap 0.15–0.30). This is *not* an independent conservation classifier, for three reasons. First, the perfect-prediction ceiling $\mathrm{adv} = 1 - 1/K$ is reached by a *dissipative* flow—Halvorsen ($a_{\mathrm{trans}} = 1.000$ over 327 changes)—and, trivially, by the periodic harmonic con-

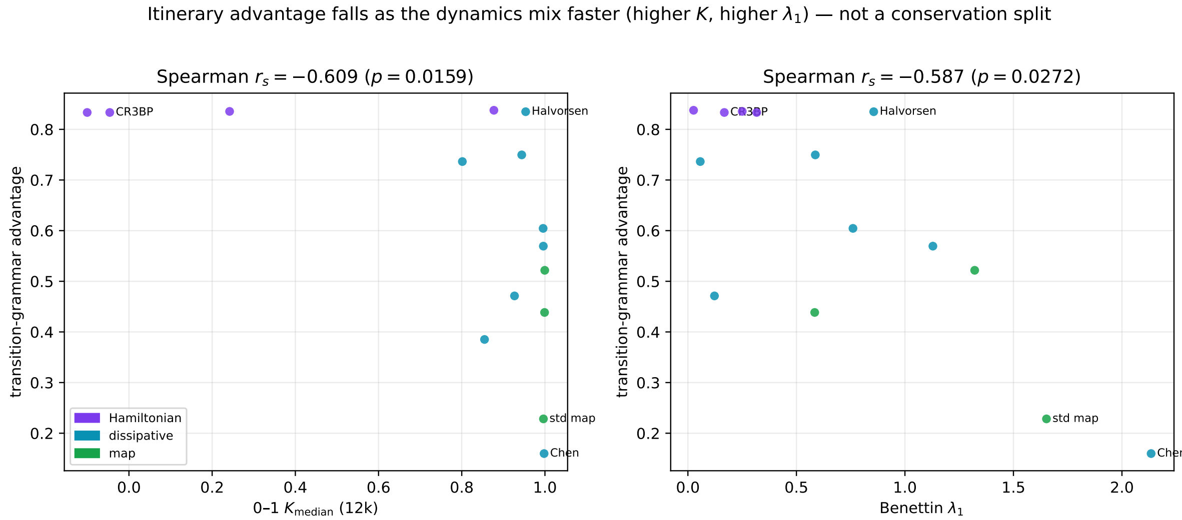

Figure 7: Held-out itinerary transition-grammar advantage versus the two mixing proxies—the 0–1 K_{median} (left) and the Benettin λ_1 (right)—over the reliable chaotic systems. The advantage falls as either proxy rises (Spearman −0.61 and −0.59) and does *not* split by family: dissipative Halvorsen reaches the Hamiltonian ceiling, while dissipative Chen and the standard map sit lowest.

trol; perfect coarse prediction is not a conservation signature. Second, the 100-change gate is a survivorship filter: it discards exactly the strongly chaotic Hamiltonians whose coarse itinerary is disordered (double pendulum adv = 0.20, Yang–Mills 0.12) because they change cells too rarely to measure, inflating the surviving Hamiltonian mean. Third, the advantage anti-correlates with both mixing proxies (Fig. 7): the 0–1 statistic (Spearman $r_s = -0.61$, $p = 0.016$, unchanged under a second independent K estimator) and the Benettin Lyapunov exponent ($r_s = -0.59$, $p = 0.03$). It is largest exactly where the dynamics mix slowest—low finite-time K and low λ_1 together. The itinerary advantage is thus another face of slow coarse-grained mixing, not orthogonal evidence of conservation.

The unifying axis is mixing rate. Weakly chaotic, dwell-dominated systems trace a nearly periodic coarse itinerary and saturate the ceiling, whether Hamiltonian (CR3BP, Pullen–Edmonds, quartic, Hénon–Heiles) or dissipative (Halvorsen); strongly mixing systems trace a disordered itinerary and score low, whether dissipative (Chen, 0.16), a map (standard map, 0.23), or Hamiltonian (double pendulum, Yang–Mills). As with the 0–1 test, coarse predictability measures how fast the chosen observable mixes, not whether phase-space volume is conserved. The picture is robust to the k-means symbolization: across 8 cell seeds the advantage–K Spearman stays in $[-0.70, -0.56]$ and the Hamiltonian mean exceeds the dissipative mean in every seed (`results/chaos_kmeans_robustness.json`). Data: `results/chaos_itinerary_advantage.json`.

3.8 Decay law versus decay length

A natural refinement asks not how *fast* correlations decay but in what *form*: a heavy power-law tail (sticky, intermittent dynamics) versus a short exponential one (hyperbolic mixing). Sticky tails are classically tied to mixed-phase-space Hamiltonians, making the decay law a candidate classifier. We test it directly: for each chaotic system we ensemble-average the normalized autocorrelation $\rho(k)$ of the first coordinate over four perturbed initial conditions and fit, over the resolvable initial decay ($\rho \geq 0.05$), both $\rho = A\,e^{-k/\tau}$ and $\rho = A\,k^{-p}$, comparing R^2. The law does not separate the

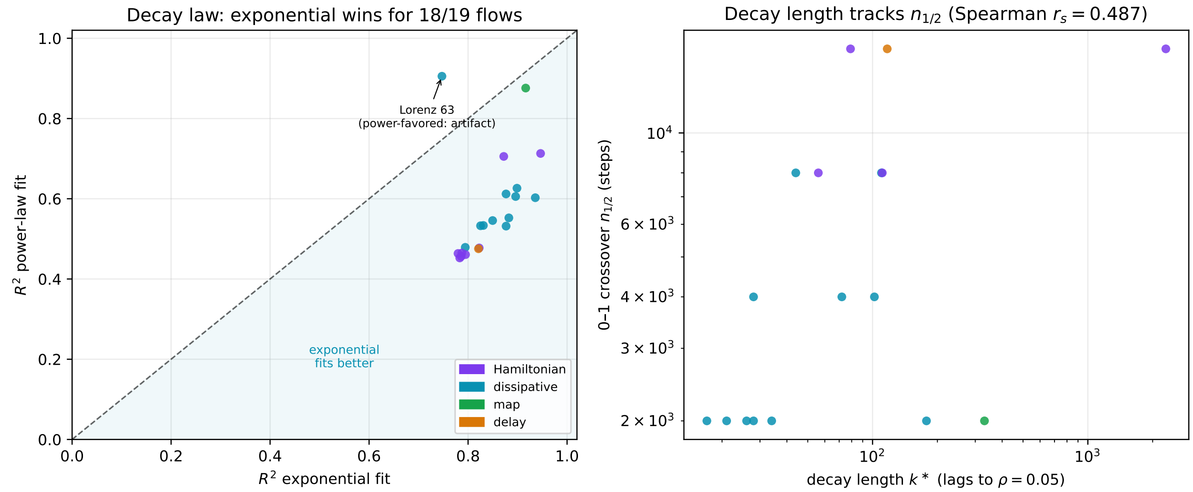

Figure 8: Left: power-law versus exponential fit quality (R^2) for the resolvable initial autocorrelation decay; nearly all flows fall below the diagonal (exponential wins), the lone exception being Lorenz-63—a textbook exponential mixer—whose marginal power-favored fit is a stable artifact of the short, oscillation-confounded window (power-favored in all 10 frequency-seed sets), not a heavy tail. Right: the decay length k^* tracks the 0–1 crossover $n_{1/2}$ across families (Spearman 0.49). The decay *law* does not classify; the decay *length* does, and it is conservation-blind.

classes: the power-favored fraction is 0 of 8 Hamiltonians and 1 of 11 dissipative flows, and that lone power-favored fit lands on Lorenz-63—the textbook *exponentially*-mixing attractor—at a small margin (+0.16 in R^2). This mislabelling is *stable*, not sampling noise: across 10 frequency-seed sets Lorenz-63 is power-favored in all 10, while only two other systems are ever power-favored (each in 1 of 10). The failure is systematic—the genuine far tail lives below $\rho = 0.05$ in the $O(1/\sqrt{MN})$ floor, and over the short resolvable window the oscillatory $\rho(k)$ of spiral flows is fit marginally better by a power law—so the exp-vs-power label is not a valid tail diagnostic. What *is* robust is the decay *length* k^*, the lags to reach $\rho = 0.05$ (Fig. 8): it reproduces the lab mixing time almost exactly (Spearman $r_s = 0.99$) and tracks the 0–1 crossover $n_{1/2}$ ($r_s = 0.49$, $p = 0.047$). The longest-memory systems—Yang–Mills ($k^* = 2515$), coupled Duffing (2303), and the standard map (331)—span the Hamiltonian and map families. Once again it is the conservation-blind mixing rate, not the decay form or the conservation structure, that the 0–1 test responds to. Data: `results/chaos_decay_law.json`.

3.9 Itinerary memory depth: order-1 is not the whole story

How much memory does the coarse itinerary carry? On the raw symbol stream the question is unanswerable for flows: dwell time dominates, so order-1, order-2, and the predict-stay baseline coincide exactly (§3.7). The grammar lives in the *route*—the run-length-encoded sequence of distinct cells visited. We collapse each held-out symbol stream to its route and compare Markov orders 1, 2, 3 (each backing off to lower orders) on the 18 chaotic systems with at least 60 held-out route steps.

Memory depth is heterogeneous and conservation-blind (Fig. 9). Half the systems (8 of 18) gain from order-2, several substantially—Mackey–Glass ($0.61 \to 0.94$), the standard map ($0.51 \to 0.76$), the spring pendulum ($0.81 \to 0.97$), and Chen, which keeps improving through order-3 ($0.58 \to 0.73 \to 0.80$)—while the rest are order-1 Markov, including four Hamiltonians already at perfect

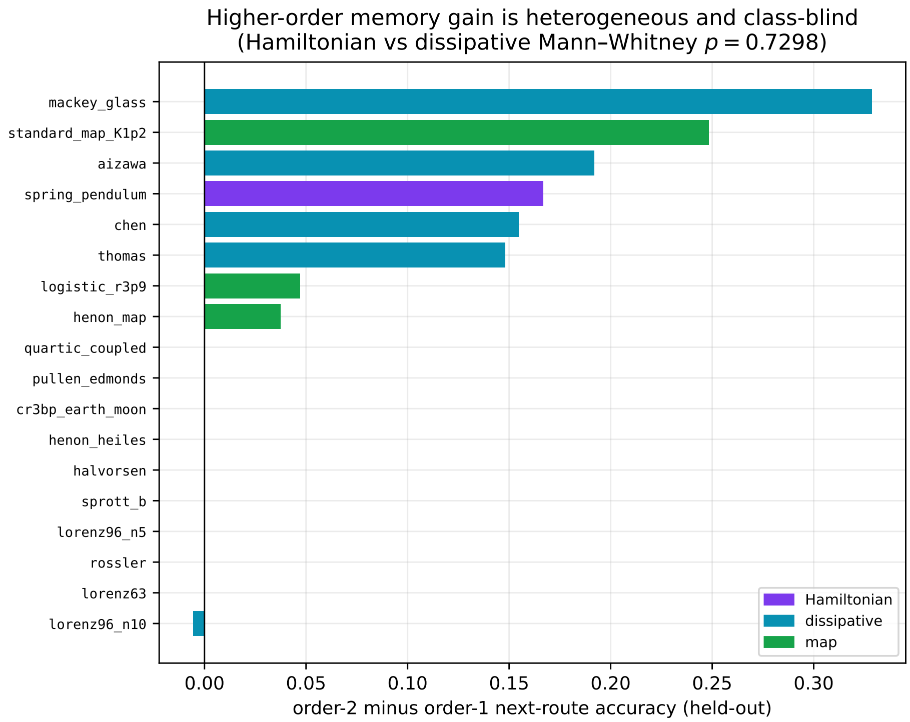

Figure 9: Order-2 minus order-1 held-out next-route accuracy for the reliable systems. The memory gain is heterogeneous and class-blind: the largest gains span dissipative (Mackey–Glass, Aizawa, Chen), map (standard map, logistic), and Hamiltonian (spring pendulum) systems, and the Hamiltonian-versus-dissipative difference is not significant (Mann–Whitney $p = 0.73$).

order-1 route accuracy (1.000). The order-2 gain does not separate the classes: mean 0.08 (dissipative) versus 0.03 (Hamiltonian), Mann–Whitney $p = 0.73$, with the largest gains spanning dissipative, map, and Hamiltonian families alike. As with every diagnostic in this section, what varies across systems is coarse dynamical structure—here the route's memory length—not the conservation of phase-space volume. This is robust to the k-means seed: across 8 seeds dissipative systems always carry the larger gain and the Hamiltonian-vs-dissipative test is non-significant in 7 of 8 ($p \in [0.045, 0.73]$); the lone marginal seed still favors dissipative, so no conservation split emerges (`results/chaos_kmeans_robustness.json`). Data: `results/chaos_markov_order.json`.

4 Parameter sweeps and stress checks

Parameter sweeps stress-test the two conclusions (Fig. 10). For Lorenz63, $\nabla \cdot f = -\sigma - 1 - \beta$ is independent of ρ; the measured $\Sigma\lambda$ remains pinned at -13.667 across $\rho \in [28, 150]$ while the largest Lyapunov exponent varies by a factor of over 33. Thus volume contraction measures dissipative structure, not chaos strength. The 0–1 test, by contrast, correctly flags non-chaotic regimes within a family: a Lorenz periodic window and a standard-map KAM regime produce low K. This supports the corrected interpretation: 0–1 is a chaos detector whose finite-time behavior depends on mixing, while $\Sigma\lambda$ is the conservative/dissipative discriminator when the vector field is available.

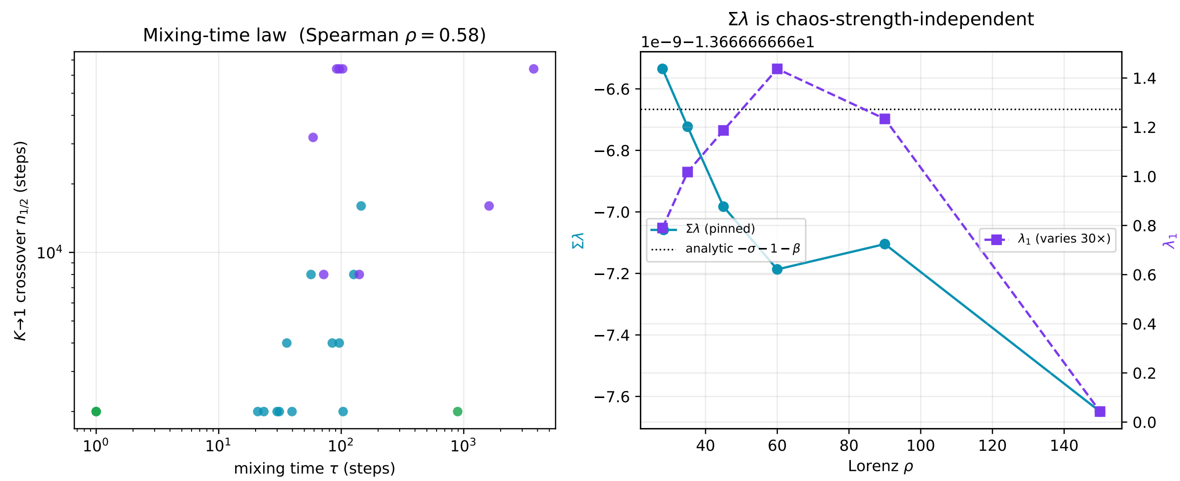

Figure 10: Left: mixing-time law, with longer observable autocorrelation associated with later K crossover. Right: Lorenz volume contraction remains pinned to the analytic trace while λ_1 changes substantially across the parameter sweep.

5 Discussion

The practical message is simple. If one only has a scalar time series and wants to ask whether the dynamics are chaotic, the 0–1 test remains useful. If one has a model vector field and wants to distinguish conservative from dissipative chaos, volume contraction is the theory-grounded diagnostic. The failure mode arises when a finite-time scalar chaos detector is quietly promoted into a structural classifier. Slow mixing then masquerades as conservative structure.

A striking secondary illustration of this failure: the quasiperiodic 2-torus control (Table 1) achieves $K = 0.961$ at 4000 steps—higher than five of the eight chaotic Hamiltonian systems. Quasiperiodic orbits produce $M_c(n) \sim n^2$ (ballistic growth) near resonance, which the log–log form of K_c records as near-one. This is not chaos detection; it is a false positive from a non-chaotic near-resonant system. Extended to 12,000 steps the same system yields $K = -0.06$ (Section 3.5), confirming that the high value is a finite-trajectory artefact: the n^2 transient saturates before the longer run ends. The result reinforces the recommendation to use K as a *chaos detector*, not as a structural classifier.

When the model vector field is unavailable—purely observational scalar time series—distinguishing conservative from dissipative chaos without computing $\Sigma\lambda$ remains an open problem. The Green–Kubo framework (Section 3.3) characterizes why the 0–1 statistic fails, but does not resolve it.

The itinerary result points in a different direction. It does not overcome the Lyapunov wall for trajectories; it changes the prediction target. Local microstate forecasts fail at long horizons, but coarse symbolic route grammar can remain predictable on held-out data. That distinction is useful for scientific machine learning systems because it separates impossible weather-like prediction from bounded climate-like structure.

A notable pattern within the itinerary data: Hamiltonian systems achieve a higher mean held-out transition accuracy (0.841, $n = 8$) than dissipative flows (0.746, $n = 10$). This is the opposite of the naive expectation that strange attractors (with their lower-dimensional structure) would be easier to learn. The likely explanation is that several Hamiltonian systems in the benchmark are semi-ergodic: they orbit near KAM tori rather than filling the energy surface uniformly, producing highly predictable transition sequences. Four of the eight reach perfect held-out accuracy (1.000).

This suggests that the grammar learnability correlates more strongly with the regularity of the attractor's coarse geometry than with the sign of $\Sigma\lambda$. Section 3.7 makes this precise: the edge is *not* an independent conservation classifier—a dissipative flow reaches the same ceiling, the gate that makes the comparison reliable discards the strongly chaotic Hamiltonians, and the advantage anti-correlates with K ($r_s = -0.61$)—so it too tracks mixing rate, not conservation.

An additional note on the random walk control: Brownian motion (the position process) gives $K = 0.045$ because the non-stationarity of the position observable causes $M_c(n)$ to grow anomalously slowly under the log–log measure. The 0–1 test correctly identifies this as non-chaotic, but for a mechanically different reason than a bounded regular orbit.

6 Limitations and claim boundary

Supported claims are limited to the 26-system benchmark and the deterministic artifacts listed in Section 7. The positive discriminator claim is empirical over 8 Hamiltonian systems and 11 dissipative finite-dimensional flows with defined vector fields; Liouville's theorem supplies the theoretical reason for Hamiltonian zero contraction, but this paper does not prove a new theorem. The cautionary 0–1 claim is stronger in practice: the observed length dependence and Green–Kubo prediction show why the statistic should not be used as a conservative/dissipative classifier. The itinerary result is held-out symbolic transition prediction, not microstate forecasting and not a claim about general trajectory prediction.

Retracted: an earlier version of this analysis incorrectly concluded that conservative chaos has $K \approx 0$; that reading is superseded here. Not supported: using the 0–1 statistic as a conservative/dissipative classifier; generalizing the benchmark to all chaotic systems without further validation; claiming a new theorem; or claiming long-horizon chaotic trajectory prediction. The manuscript intentionally preserves the self-disproved 0–1 result and the corrected claim boundary rather than deleting the error.

7 Reproducibility and data availability

All reported numbers are read from deterministic, seeded artifacts. Data and figure-generation scripts are available at `https://github.com/MonongahelaHellbender/chaos-universality-paper`. The primary evidence files are:

> `results/chaos_universality_lab_best.json`;
> `results/chaos_mixing_time_analysis.json`; `results/chaos_green_kubo.json`;
> `results/chaos_parameter_sweep.json`; `results/chaos_itinerary_prediction.json`;
> `results/chaos_slope_analysis.json`; `results/chaos_observable_selection.json`;
> `results/chaos_itinerary_advantage.json`; `results/chaos_decay_law.json`;
> `results/chaos_run_length.json`; `results/chaos_markov_order.json`; and
> `results/chaos_kmeans_robustness.json`.

Figures are regenerated by `scripts/chaos_make_figures.py` (main results, Figs. 1–5) and `scripts/chaos_extension_figures.py` (diagnostic battery, Figs. 5–9). The slope analysis (Table 2, §3.5), observable screening (§3.6), itinerary-advantage meta-analysis (§3.7), decay-law analysis (§3.8), and order-k route-grammar analysis (§3.9) are regenerated by the like-named scripts in `scripts/` (the decay-law and order-k scripts import the system definitions from `chaos_slope_analysis.py`). The k-means seed-robustness of the §3.7 and §3.9 verdicts is checked by `scripts/chaos_kmeans_robustness.py`.

A Effective-sample interpretation of the 0–1 crossover

Equation (4) implies that K reaches its asymptotic chaotic value only when the trajectory is long relative to the observable's integrated autocorrelation time. The artifacts estimate this time by a Sokal-style windowed autocorrelation procedure [12]. Across the cross-system ladder, Spearman $r_s(\tau, n_{1/2}) = 0.58$ ($t = 3.0$, df $= 17$, $p < 0.01$; $n = 19$ chaotic systems with measurable crossover; 4 Hamiltonian systems whose K never exceeded 0.5 within the audit range are excluded): longer observable mixing time delays the first length where $K > 0.5$. Dissipative systems have median crossover near 3k steps, while Hamiltonian systems have median crossover near 12k steps. The relevant variable is therefore effective sample size after decorrelation, not conservation versus dissipation.

This interpretation is operational. The same closed form gives a run-length calculator: from the observable autocovariance $C(k)$ it returns the smallest decimated sample count at which the predicted K crosses a target threshold (multiply by the decimation stride for raw steps). Driven by the measured autocovariance it reproduces the ladder crossover for 14 of 17 systems exactly and 17 of 17 within one ladder step; on a controlled AR(1) observable $C(k) = \rho^k$ the returned length grows with $\tau_{\text{int}} = (1 + \rho)/(1 - \rho)$, rising from 30 to 500 samples as ρ runs $0.5 \to 0.98$, as required. When only a single scalar is known, the one-parameter rule $N \approx A\,\tau_{\text{int}}$ with $A \approx 65$ (calibrated on the 16 benchmark flows with a measurable crossover) gives an order-of-magnitude estimate—median fold-error 1.5×, 88% within 3×, worst case 6.6× (the coupled Duffing oscillator)—and applies to continuous-time flows only: maps decorrelate in a single step, so $\tau \approx 1$ carries no length information. The calculator is `scripts/chaos_run_length.py`.

B Reference table for the main claim

Question	Correct diagnostic	Boundary
Is the scalar series chaotic?	0–1 K	Requires sufficient effective length
Is the flow conservative or dissipative?	$\Sigma\lambda = \langle \nabla \cdot f \rangle$	Requires vector field/model
Can the exact future state be predicted?	No long-horizon claim	Lyapunov wall remains
Can coarse route grammar be predicted?	Held-out itinerary model	Symbolic transitions only

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Table 1: The 26-system benchmark. λ_1: largest Lyapunov exponent (positive = chaotic); "n.c." means not computed via global algorithm (Mackey–Glass requires delay-embedding; known value from the literature is $\lambda_1 \approx 0.007$ for $\tau = 17$). K: 0–1 statistic for the primary initial condition, evaluated on the 0–1 test's extended trajectory ($3\times$ the base $n_{\text{steps,flow}}$, $N \approx 12{,}000$ steps for the flows); for slow-mixing Hamiltonians K is strongly length-dependent, so this value sits well above the short-run reading (double pendulum: $K = -0.13$ at 4,000 steps, mean over four ICs, climbing toward one by 12–16k; see Fig. 1). $\Sigma\lambda$: measured phase-space volume contraction (N/A for maps and the delay system, which lack a finite-dimensional smooth vector field).

System	Type	Dim	λ_1	K	$\Sigma\lambda$
Dissipative flows					
Lorenz 63	Dissipative	3	+0.761	0.996	−13.667
Rössler	Dissipative	3	+0.057	0.801	−5.737
Duffing (forced)	Dissipative	3	+0.114	0.729	−0.300
Lorenz 96 ($N = 5$)	Dissipative	5	+0.587	0.944	−5.000
Aizawa	Dissipative	3	+0.122	0.927	−0.183
Sprott B	Dissipative	3	+0.217	0.999	−1.000
Halvorsen	Dissipative	3	+0.856	0.954	−4.200
Thomas	Dissipative	3	+0.046	0.995	−0.625
Chen	Dissipative	3	+2.134	0.998	−10.000
Ueda (forced)	Dissipative	3	+0.100	0.513	−0.050
Lorenz 96 ($N = 10$)	Dissipative	10	+1.129	0.996	−10.000
Hamiltonian flows					
Double pendulum	Hamiltonian	4	+0.888	0.770	0.000
Hénon–Heiles	Hamiltonian	4	+0.026	0.877	0.000
CR3BP (Earth–Moon)	Hamiltonian	4	+0.168	−0.046	0.000
Yang–Mills (x^2y^2)	Hamiltonian	4	+0.241	−0.110	0.000
Pullen–Edmonds	Hamiltonian	4	+0.317	−0.100	0.000
Quartic coupled	Hamiltonian	4	+0.250	0.242	0.000
Spring pendulum	Hamiltonian	4	+0.116	0.979	0.000
Coupled Duffing (Ham.)	Hamiltonian	4	+0.214	0.957	0.000
Maps (no vector field; $\Sigma\lambda$ not applicable)					
Logistic ($r = 3.9$)	Map	1	+1.322	1.000	N/A
Hénon map	Map	2	+0.584	0.999	N/A
Standard map ($\kappa = 1.2$)	Map	2	+1.652	0.996	N/A
Delay system ($\Sigma\lambda$ not applicable)					
Mackey–Glass ($\tau = 17$)	Delay	∞	n.c.†	0.855	N/A
Non-chaotic controls					
Harmonic oscillator	Control	2	0.000	0.599	N/A
Quasiperiodic 2-torus	Control	2	0.000	0.961	N/A
Random walk	Control	2	—	0.045	N/A

Table 2: Growth-rate exponent α at $N = 12{,}000$ steps. α: full-range log–log slope of $M_c(n)$; α_{lo}: long-range ($n \geq N_{\mathrm{dec}}/2$) sub-slope; K: median 0–1 statistic over 12 random frequency seeds at this length (the main-lab pipeline of Table 1 can differ for slow-mixing Hamiltonians; a single 12-frequency draw is c-sampling unstable for these systems—see text and the `K_seed` fields of the JSON for per-system ranges). Negative α_{lo} signals M_c saturation. Bold: the two systems with $\alpha \approx 1$ yet a robustly low K (the α–K decoupling). Full results in `results/chaos_slope_analysis.json`.

System	Family	α	α_{lo}	K
Dissipative flows				
Lorenz 63	Dissipative	0.929	+0.934	+0.999
Rössler	Dissipative	0.615	−0.254	+0.712
Duffing (forced)	Dissipative	0.916	+0.995	+0.322
Lorenz 96 ($N = 5$)	Dissipative	0.884	+0.090	+0.956
Aizawa	Dissipative	0.853	+0.191	+0.686
Sprott B	Dissipative	0.956	+0.969	+0.998
Halvorsen	Dissipative	1.018	+1.331	+0.964
Thomas	Dissipative	1.098	+1.132	+0.996
Chen	Dissipative	1.024	+1.103	+0.994
Ueda (forced)	Dissipative	0.971	+1.297	+0.347
Lorenz 96 ($N = 10$)	Dissipative	0.902	+0.978	+0.996
Hamiltonian flows				
Double pendulum	Hamiltonian	1.312	+1.296	+0.709
Hénon–Heiles	Hamiltonian	0.949	+1.111	+0.577
CR3BP (Earth–Moon)	Hamiltonian	**0.847**	+0.887	**−0.046**
Yang–Mills (x^2y^2)	Hamiltonian	−0.083	+0.275	+0.069
Pullen–Edmonds	Hamiltonian	0.694	−0.149	−0.049
Quartic coupled	Hamiltonian	**0.888**	+1.061	**−0.108**
Spring pendulum	Hamiltonian	0.972	+0.924	+0.320
Coupled Duffing (Ham.)	Hamiltonian	0.759	+0.912	+0.942
Maps				
Logistic ($r = 3.9$)	Map	1.011	+1.107	+0.999
Hénon map	Map	1.020	+1.008	+1.000
Standard map	Map	1.051	+1.129	+0.998
Delay system				
Mackey–Glass	Delay	1.400	+1.539	+0.733
Non-chaotic controls				
Harmonic oscillator	Control	0.543	−0.806	−0.090
Quasiperiodic 2-torus	Control	0.906	+0.373	+0.097
Random walk	Control	0.008	+0.162	+0.157